

Neonatal Brain Volumes and Early Parenting Behavior as Mediators in Associations Between Prenatal Social Disadvantage and Socioemotional Problems in Toddlers

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ABSTRACT

BACKGROUND: Social disadvantage has been associated with early socioemotional difficulties. In this study, we examined mechanisms that relate prenatal social disadvantage (PSD) to the development of early socioemotional problems by testing whether these associations were mediated by 1) neonatal brain volumes (BVs) and/or 2) early parenting behaviors.

METHODS: Women were recruited early in their pregnancies and followed prospectively. PSD encompassed access to material (e.g., income-to-needs, health insurance, area deprivation, nutrition, education) resources during pregnancy. Shortly after birth, neonates underwent structural magnetic resonance scanning. Mother-child dyads returned for parenting observations at child age 1 year, and parents reported child socioemotional problems (Infant-Toddler Social and Emotional Assessment: externalizing, dysregulation, internalizing) at age 2 years ($N = 267$; 45% female). Simple and parallel mediation models were used to test hypotheses.

RESULTS: Greater PSD was associated with increased externalizing and dysregulation symptoms at age 2 years. PSD-associated reductions in neonatal BVs (cortical gray matter, white matter, total brain) mediated both PSD-externalizing and PSD-dysregulation associations. The PSD-externalizing association was additionally mediated by early parenting behaviors, particularly nonsupportive parenting behaviors. Thus, for externalizing symptoms, nonsupportive parenting behaviors and mediating brain metrics were examined simultaneously in parallel mediation models. Nonsupportive parenting remained a significant mediator, while neonatal BVs were no longer significant.

CONCLUSIONS: PSD-associated brain structural alterations at birth may serve as early risk factors for the development of multidimensional socioemotional difficulties in toddlerhood. However, parenting emerged as a stronger mediator for externalizing problems, lending support to parenting behaviors as key intervention targets for the prevention of externalizing problems during early childhood.

<https://doi.org/10.1016/j.biopsych.2025.05.015>

Early socioemotional skills—or “emerging abilities to experience, regulate, and express emotions in socially- and culturally-appropriate ways” (1,2)—are critical for children’s academic, social, and emotional well-being. Difficulties in early socioemotional development [e.g., subclinical deficits in these skills relative to same-aged peers (1)] are strong predictors of later socioemotional problems and higher incidences of clinical psychiatric disorders (3–7). Moreover, socioemotional difficulties tend to emerge early and worsen with age (8,9), making early manifestations of difficulties particularly important signals that highlight opportunities for early intervention and prevention. Children who grow up with high levels of social disadvantage, defined as lacking access to material resources, have disproportionately higher rates of socioemotional difficulties (10–14). Meta-analytic estimates suggest that children

exposed to high levels of social disadvantage are up to 3 times more likely to develop emotional or behavioral problems than their peers exposed to lower levels of social disadvantage (13,14). Given the importance and predictive value of early socioemotional difficulties for later psychopathology, the current project aims to understand mechanisms through which the earliest exposure to social disadvantage, prenatal social disadvantage (PSD), is associated with risk for early socioemotional dysfunction. The prenatal period is particularly important as fetal development serves as a basis of future health and disease, including in the socioemotional domain (15–17).

There may be multiple mechanisms through which PSD influences socioemotional outcomes, including via the impact of PSD on fetal brain development. Greater PSD has been

associated with smaller global and regional brain volumes (BVs) (18), aberrant white matter (WM) microstructure (19), and altered resting-state functional connectivity (20) in neonates. Across development, there is evidence that such neural alterations not only correlate with socioemotional abilities (21–27) but also mediate the link between disadvantage and socioemotional abilities in later childhood and adolescence (28,29). Thus, it is possible that these neural phenotypes near birth similarly mediate the relationship between PSD and early socioemotional abilities. To date, one study (30) has found that the relationship between social disadvantage at birth and greater externalizing symptoms in offspring at age 2 years was mediated by stronger neonatal striatum-frontopolar functional connectivity. However, whether PSD-associated structural brain alterations at birth similarly mediate the link to socioemotional problems has not yet been investigated. If such an association could be detected, it might identify neonatal brain volumetric alterations as early biomarkers of risk for future socioemotional problems.

Parenting behaviors may constitute another candidate mechanism, as social disadvantage and socioemotional outcomes have both been linked to parenting (31–38). Parenting behaviors are shaped by the broader environmental context, and certain aspects of living in disadvantage (e.g., increased stress, increased demand for vigilance, diminished access to resources, increased time spent working) influence the relative expression and types of parenting practices (39–41). PSD and postnatal social disadvantage have been associated with variation in both supportive (e.g., responsivity) and non-supportive (e.g., harsh control) types of parenting behaviors (38,42). In turn, parental responsivity and harsh control have been linked to child socioemotional competence (43,44). For example, high levels of responsivity or warmth have been associated with lower levels of childhood internalizing and externalizing symptoms (35,36), while high levels of harsh parenting behaviors have been associated with higher levels of childhood internalizing and externalizing symptoms (35). In older children, both supportive and non-supportive parenting behaviors have mediated associations between social disadvantage and multiple childhood socioemotional problems [e.g., externalizing (35,43,45), internalizing (46), dysregulation (12)]. These pathways are comparatively less investigated early in life (45,47); however, if these associations exist as early as the first years of life, it could support the importance of earlier prevention or intervention for optimal socioemotional development.

Critically, it is unlikely that there is only one single mediator that explains the relationship between PSD and early socioemotional outcomes. Thus, a comprehensive model should investigate whether different mediators account for independent variance, whether one mediator may be more powerful than another, and, more exploratorily, whether mediators interact with each other. Better understanding the relative contributions of different mediators could inform where to direct efforts (e.g., which mechanistic target) for maximal benefit in socioemotional development, especially in resource-limited contexts.

To address these questions about mediators of the relationships between PSD and early socioemotional abilities, we leveraged a prospective longitudinal study that began

prenatally and followed participants through offspring age 2 years. We tested the following hypotheses: 1) PSD would be associated with increased risk for impairments in socioemotional development (more externalizing, dysregulation, and internalizing); these associations would be mediated by 2) neonatal BVs and 3) supportive and non-supportive parenting behaviors; and 4) neonatal BVs and parenting behaviors would make independent contributions in mediating associations (e.g., would both be significant mediators in parallel mediation models) between PSD and offspring socioemotional outcomes.

METHODS AND MATERIALS

Participants and Procedures

As previously described, the eLABE (Early Life Adversity Biological Embedding) study (48,49) recruited a prospective sample of 398 mother-child dyads enriched for exposure to poverty. Briefly, pregnant women were enrolled early in their pregnancy and completed assessments during each trimester. Shortly after birth (postmenstrual age at scan: mean = 41.0 weeks, SD = 1.5), neonates underwent magnetic resonance imaging (MRI) scans. Mother-child dyads returned for follow-up assessments at child ages 1 and 2 years. Written informed consent was obtained from mothers prior to participation. All protocols were approved by the Institutional Review Board at Washington University in St. Louis.

Measures

Prenatal Social Disadvantage. Consistent with previous investigations in this sample, PSD was modeled with a latent construct to represent this complex multifaceted theoretical construct. As previously described (48), PSD is based on highly related indicators of material resources during pregnancy: income-to-needs ratio, Area Deprivation Index, health insurance status, maternal education, and maternal nutrition. PSD scores range from -2.2 to $+1.5$, where higher scores represent greater disadvantage. See the [Supplement, section 1](#) for more details for indicators, correlations, and construction of the PSD measure. Measuring social disadvantage during the prenatal period enables mediation analyses with correct temporal sequence.

Infant-Toddler Social and Emotional Assessment.

The Infant-Toddler Social and Emotional Assessment (ITSEA) is a parent-reported questionnaire that measures socioemotional problems and behavioral competencies in children ages 12 to 36 months (50). This study used the externalizing, internalizing, and dysregulation domains reported at the 2-year follow-up. Externalizing behaviors include impulsivity, aggression, and defiance. Internalizing behaviors encompass depression, anxiety, and shyness. Dysregulation behaviors encompass problems with sleeping, eating, and regulating negative emotions. All analyses used age- and sex-normed T scores (mean = 50, SD = 10) (51). Higher scores represent greater socioemotional problems.

Parent-Child Interaction Rating Scales. An adapted version of the Parent-Child Interaction Rating Scales (52) was

used to code parent-child interactions across 3 videotaped tasks. Raters were trained to reliability. Parenting behaviors were scored on a 7-point scale and averaged across tasks. As previously described (42), the supportive parenting behaviors composite ($\alpha = 0.861$) averaged scores for sensitivity and positive regard. The nonsupportive parenting behaviors composite ($\alpha = 0.726$) averaged scores for intrusiveness, detachment, and negative regard. Higher scores represent greater intensity and frequency of the observed behavior. Analyses used parenting behaviors at year 1 to enable mediation analyses with temporal precedence of parenting (year 1) relative to socioemotional abilities (year 2). See the [Supplement, section 2](#) for more details and reliabilities.

Magnetic Resonance Imaging. MRI data collection and processing were conducted as previously described (18) and are further detailed in the [Supplement](#). Briefly, all infants were scanned during natural sleep without sedation during the first few weeks of life (mean = 3.17 weeks). Prior work in this sample found associations between greater PSD and volumetric reductions in 6 neonatal brain regions: total brain, cortical gray matter (GM), subcortical GM, WM, and cerebellar volumes, and mean gyrification index (GI) (18). These 6 brain regions were the *a priori* metrics of interest.

Data Analysis

PSD and Child Outcomes. Predictors were added to both linear and nonlinear models to assess fit; however, no analyses indicated that nonlinear fits were better ([Supplement, section 4](#)). Therefore, only linear models are reported below. ITSEA outcomes that were related to PSD were investigated in all subsequent analyses.

Mediations by Neonatal Brain Metrics. Separate regressions tested whether 1) each of the 6 *a priori* brain metrics related to outcomes and 2) related brain metrics mediated the relationships between PSD and outcomes [*mediation* package version 4.5.0 (53) in R].

Mediations by Parenting Behaviors. Separate regressions tested whether 1) PSD related to parenting behaviors, 2) parenting behaviors related to outcomes, and 3) parenting behaviors mediated the relationships between PSD and outcomes [*mediation* package version 4.5.0 (53)]. Analyses were conducted separately for supportive and nonsupportive parenting behaviors in R.

Parallel Mediations. Parallel mediation models were used when a PSD-outcome association was mediated by 1) both supportive and nonsupportive parenting behaviors and/or 2) both neonatal brain metrics and parenting behaviors. Parallel mediations were conducted using SAS version 9.4 (SAS Institute Inc.).

Exploratory Interactions Between Mediators. Interaction terms were added to regressions to investigate whether interactions between 1) supportive and nonsupportive parenting, 2) supportive parenting and each neonatal brain metric, and 3) nonsupportive parenting and each neonatal

brain metric predicted socioemotional abilities. Moderated mediations examined whether the strength of mediation by one variable depended on the level of another variable.

Covariates. We tested covariates potentially causally related to the outcomes of interest (54,55). For models with ITSEA domains as outcomes, none of the examined covariates (gestational age at birth, child sex, age at assessment, age at MRI scan) were related to ITSEA outcomes (all p s > .05). Therefore, no covariates were included in these models. For models with neonatal BVs as outcomes (e.g., path A in mediations by neonatal brain volumes), postmenstrual age at scan related to PSD and to neonatal BVs, and it was included as a covariate in this path.

Results for all primary analyses were corrected using false discovery rate (FDR) correction (56). For correction, tests were grouped by the research question and by the outcome variable. See the [Supplement, section 4](#) for further details.

RESULTS

At the 2-year follow-up, complete ITSEAs were obtained for 267 children. Sample characteristics are detailed in [Table 1](#). Race information is included in the demographic table for comparison to other samples in the literature, but it is not included in any analyses. Compared with mother-child dyads included in analyses, nonincluded dyads had a different distribution of health insurance types ($n = 131$, $p = .046$) and higher internalizing scores ($n = 3$, $p = .042$). Excluded dyads also had higher postmenstrual ages at scan ($n = 112$, $p = .04$), lower scores for positive regard ($n = 33$, $p = .039$; although not different in the supportive parenting composite), and a different distribution of child race ($n = 131$, $p = .042$). See the [Supplement, section 5](#) for more details about sample characteristics, both for the recruitment sample ([Table S3](#)) and included versus excluded dyads ([Table S4](#)).

PSD and Socioemotional Problems

Greater PSD was associated with greater externalizing ($\beta = 0.28$, $q < .0001$) and dysregulation ($\beta = 0.22$, $q < .001$) symptoms at age 2. PSD was not significantly related to internalizing symptoms after FDR correction ($\beta = 0.10$, $q = .11$) ([Figure 1](#)).

Mediations by Neonatal Brain Metrics

Smaller volumes of cortical GM, total BV, WM, subcortical GM, cerebellum, and mean GI at birth—all of which have previously shown to relate to PSD in this sample (18)—were associated with greater externalizing symptoms at age 2 ([Table 2](#)). Similarly, reductions in each of these PSD-associated BVs, except mean GI and cerebellum, were associated with greater dysregulation symptoms at age 2 ([Table 2](#)). Smaller cortical GM, WM, and total BVs at birth each partially mediated the association between greater PSD and greater externalizing symptoms ([Figure 2A–C](#) and [Table 3](#)). However, when these 3 brain metrics were simultaneously entered into a parallel mediation model, none significantly mediated the PSD-externalizing association ([Table S5](#)), potentially because they were highly intercorrelated (r s = 0.80–0.95). Smaller cortical GM and total BV at birth also partially mediated the association between greater PSD and greater dysregulation symptoms ([Figure 2D, E](#) and [Table 3](#)).

Table 1. Sample Characteristics

Maternal Characteristics, <i>N</i> = 267	Mean (SD) [Range] or <i>n</i> (%)
Prenatal Social Disadvantage Latent Factor Score	−0.06 (0.96) [−2.22 to 1.55]
Income-to-Needs Ratio	2.80 (2.85) [0.38 to 12.48]
National Area Deprivation Index	68.70 (24.15) [7 to 100]
Healthy Eating Index	58.58 (10.12) [31.70 to 80.67]
Highest Level of Education	
<12th grade	17 (6.4%)
High school degree/GED	123 (46.1%)
College degree, 4 years	36 (13.5%)
Graduate degree	59 (22.1%)
Health Insurance Status	
Medicaid or Medicare	95 (35.5%)
Individual or group health insurance	144 (53.9%)
Uninsured	28 (10.5%)
Supportive Parenting Behaviors, <i>n</i> = 178	4.79 (0.90) [2.5 to 7]
Sensitivity	4.84 (0.99) [2.67 to 7.00]
Positive regard	4.73 (0.98) [2.00 to 7.00]
Nonsupportive Parenting Behaviors, <i>n</i> = 178	1.62 (0.49) [1.00 to 3.44]
Negative regard	1.31 (0.46) [1.00 to 3.00]
Intrusiveness	2.19 (0.91) [1.00 to 5.33]
Detachment	1.37 (0.63) [1.00 to 4.33]
Infant Characteristics, <i>N</i> = 267	
Gestational Age at Birth, Weeks	37.97 (1.92) [28 to 41]
Age at 2-Year Follow-Up, Years	2.03 (0.11) [1.91 to 2.53]
Sex Assigned at Birth	
Female	120 (45%)
Male	147 (55%)
Child Race	
African American	156 (58.4%)
Asian Indian	1 (0.38%)
Chinese	3 (1.12%)
Korean	1 (0.38%)
Multiracial ^a	1 (0.38%)
Other ^b	3 (1.12%)
White	102 (38.2%)
Postmenstrual Age at Scan, Weeks, <i>n</i> = 241	41.01 (1.49) [37 to 45]
Neonatal Brain Metrics, mm ³ , <i>n</i> = 241	
Total cortical GM	119,086.9 (15,746.36) [73,818.51 to 161,621.8]
Total cerebral WM	183,455.0 (19,685.02) [131,240.4 to 244,443.3]
Total cerebellum	27,996.71 (4425.81) [15,645.7 to 45,875.2]
Subcortical GM	26,866.08 (2798.10) [18,047.16 to 35,261.38]
Total brain volume	355,797 (39,388.11) [253,440.4 to 472,884.4]
Mean GI	1.96 (0.10) [1.53 to 2.20]
Child Outcomes	
ITSEA T Scores, <i>N</i> = 267	
Externalizing	54.01 (10.42) [33 to 99]

Table 1. Continued

Maternal Characteristics, <i>N</i> = 267	Mean (SD) [Range] or <i>n</i> (%)
Dysregulation	45.83 (11.43) [26 to 84]
Internalizing	47.74 (10.54) [27 to 77]

GED, General Educational Development; GI, gyrification index; GM, gray matter; ITSEA, Infant-Toddler Social and Emotional Assessment; WM, white matter.

^aMultiracial participants reported Chinese/White.

^bParticipants did not further specify race.

When added to a parallel mediation model, neither of these brain metrics were significant mediators (Table S5).

Mediations by Parenting Behaviors

Greater PSD was associated with fewer supportive parenting behaviors ($\beta = -0.62$, $q < .0001$) and more nonsupportive parenting behaviors ($\beta = 0.57$, $q < .0001$) (Figure S3). More supportive parenting behaviors were associated with fewer externalizing ($\beta = -0.27$, $q = .0002$) and dysregulation ($\beta = -0.18$, $q = .02$) symptoms. Conversely, more nonsupportive parenting behaviors were associated with more externalizing ($\beta = 0.37$, $q < .0001$) and dysregulation ($\beta = 0.18$, $q = .02$) symptoms (Figure S4). Supportive and nonsupportive parenting behaviors each significantly mediated the association between PSD and externalizing symptoms (Figure 3). When both types of parenting behaviors were entered into a parallel mediation model, nonsupportive parenting behaviors significantly mediated associations between PSD and externalizing behaviors, but supportive parenting behaviors did not (Figure S5). No parenting behavior mediated the association between PSD and dysregulation symptoms (Figure 3).

Relative Roles of Parenting and Brain Mediators

While only brain metrics mediated the link between PSD and dysregulation symptoms, both brain metrics and parenting behaviors mediated the relationship between PSD and externalizing symptoms. Therefore, we conducted a series of parallel mediation models to examine whether brain metrics and parenting behaviors, when entered simultaneously, accounted for unique variance in mediating the link between PSD and externalizing symptoms. When examining simultaneous contributions of nonsupportive parenting and brain metrics, nonsupportive parenting was consistently a significant mediator, while the mediation by brain metrics (cortical GM, WM, total BV) was no longer significant (Table 4). Conversely, when examining simultaneous contributions of supportive parenting behaviors and neonatal brain metrics, neither supportive parenting nor neonatal brain metrics mediated the association between PSD and externalizing symptoms (Table 4). The correlations between parenting composites and brain metrics ranged between |0.23| and |0.31|. See the Supplement for correlations between mediators (Table S6).

Exploratory Interactions Between Mediators

To understand how different mediators of the same pathways potentially interact with each other, we explored several

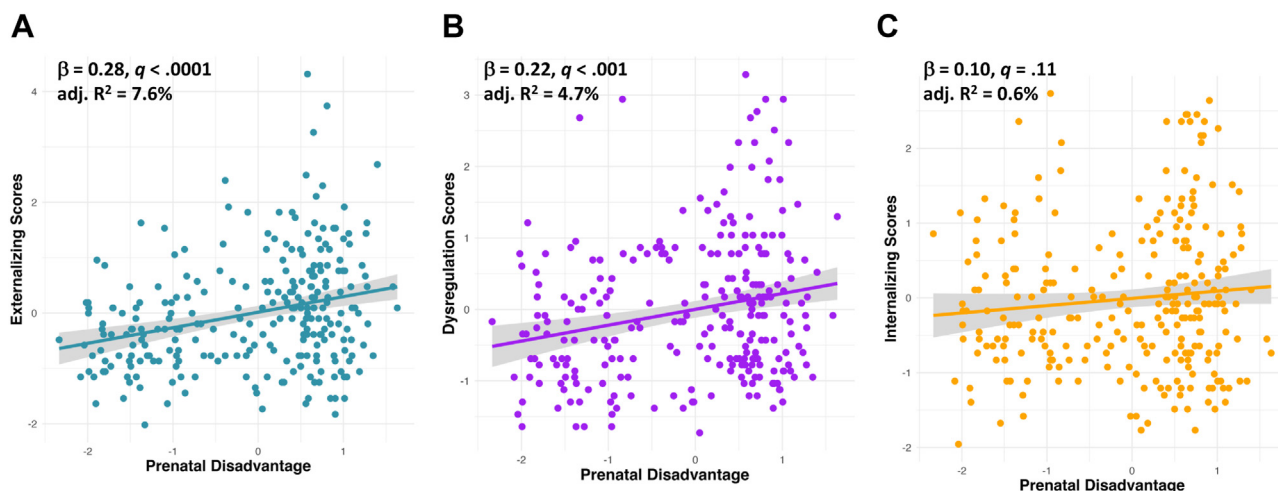


Figure 1. Scatterplots of the relationship between prenatal social disadvantage and externalizing, dysregulation, and internalizing scores at year 2. Greater prenatal social disadvantage was linearly associated with greater externalizing symptoms (blue) (A) and dysregulation (purple) symptoms (B), but not with internalizing (orange) symptoms (C) at age 2 years. Standardized estimates (β), q values (false discovery rate-adjusted p values), and adjusted (adj.) R^2 values are provided for each relationship.

interaction models. Supportive and nonsupportive parenting behaviors did not significantly interact to predict either externalizing ($p = .27$) or dysregulation ($p = .17$). Parenting behaviors and neonatal brain metrics did not significantly interact to predict either externalizing or dysregulation (see the [Supplement](#)). Finally, we used moderated mediation models to test whether the existence or strength of the mediating role of neonatal BVs in associations between PSD and socioemotional symptoms depended on the strength or type of parenting behaviors, all of which were nonsignificant ([Table S7](#)).

Follow-Up Analyses

To better understand which specific parenting behaviors were driving associations, all significant findings with parenting composites were tested with individual parenting behaviors. Sensitivity and detachment were the main drivers of supportive and nonsupportive findings, respectively. See the [Supplement](#) for detailed results ([Supplement, section 12](#) and [Tables S8–S11](#)). We

also examined whether cognition mediated PSD-socioemotional outcome associations, but all analyses were nonsignificant ([Supplement, section 13](#)).

DISCUSSION

In this longitudinal study, PSD was positively associated with greater child externalizing and dysregulation problems at age 2. Importantly, both associations were mediated by smaller neonatal BVs at birth (externalizing: cortical GM, WM, total BV; dysregulation: cortical GM, total BV). The development of externalizing symptoms was additionally mediated by supportive and nonsupportive parenting behaviors at age 1, with nonsupportive parenting being the stronger mediator. Critically, when parenting behaviors and neonatal brain mediators were entered simultaneously into parallel mediation models, only nonsupportive parenting behaviors remained significant, suggesting that parenting behaviors may be an even stronger mediator than neonatal BVs for the development of externalizing symptoms.

Taken together, the results of this study suggest that PSD-associated alterations in structural brain development in utero are meaningfully related to the development of early socioemotional problems across multiple domains—a finding that underscores the importance of ensuring that pregnant women have adequate resources to protect them from adversity to support optimal fetal brain development and simultaneously points to a need for future studies to examine more specific biological pathways (e.g., inflammation, sleep disruption) through which PSD is associated with altered structural brain development in utero. While these early brain structural alterations are related to the development of multiple domains of socioemotional problems, for the development of externalizing problems specifically, parenting behavior may be an even stronger mediator and potentially a key intervention target. Ultimately, these findings suggest that mediational pathways may vary by the specific socioemotional domain; thus, comprehensive socioemotional development would be optimally

Table 2. Associations Between Neonatal Brain Volumes and Socioemotional Outcomes at Age 2 Years

Brain Volumes at Birth	Externalizing			Dysregulation		
	β	p	q	β	p	q
Total Cortical Gray Matter	−0.35 ^a	<.0001 ^a	<.001 ^a	−0.24 ^a	<.01 ^a	.02 ^a
Total White Matter	−0.27 ^a	<.0001 ^a	<.0001 ^a	−0.18 ^a	<.01 ^a	.01 ^a
Total Cerebellum	−0.25 ^a	<.01 ^a	<.01 ^a	−0.17	.08	.09
Subcortical Gray Matter	−0.25 ^a	<.01 ^a	<.01 ^a	−0.18 ^a	.03 ^a	.04 ^a
Total Brain Volume	−0.33 ^a	<.0001 ^a	<.0001 ^a	−0.22 ^a	<.01 ^a	.01 ^a
Mean GI	−0.17 ^a	.020 ^a	.02 ^a	−0.11	.15	.15

Standardized estimates (β), p values, and q values (FDR-corrected p values) were corrected within each outcome. Postmenstrual age at scan was included as a covariate in each model.

FDR, false discovery rate; GI, gyrification index.

^aResults are significant after FDR correction.

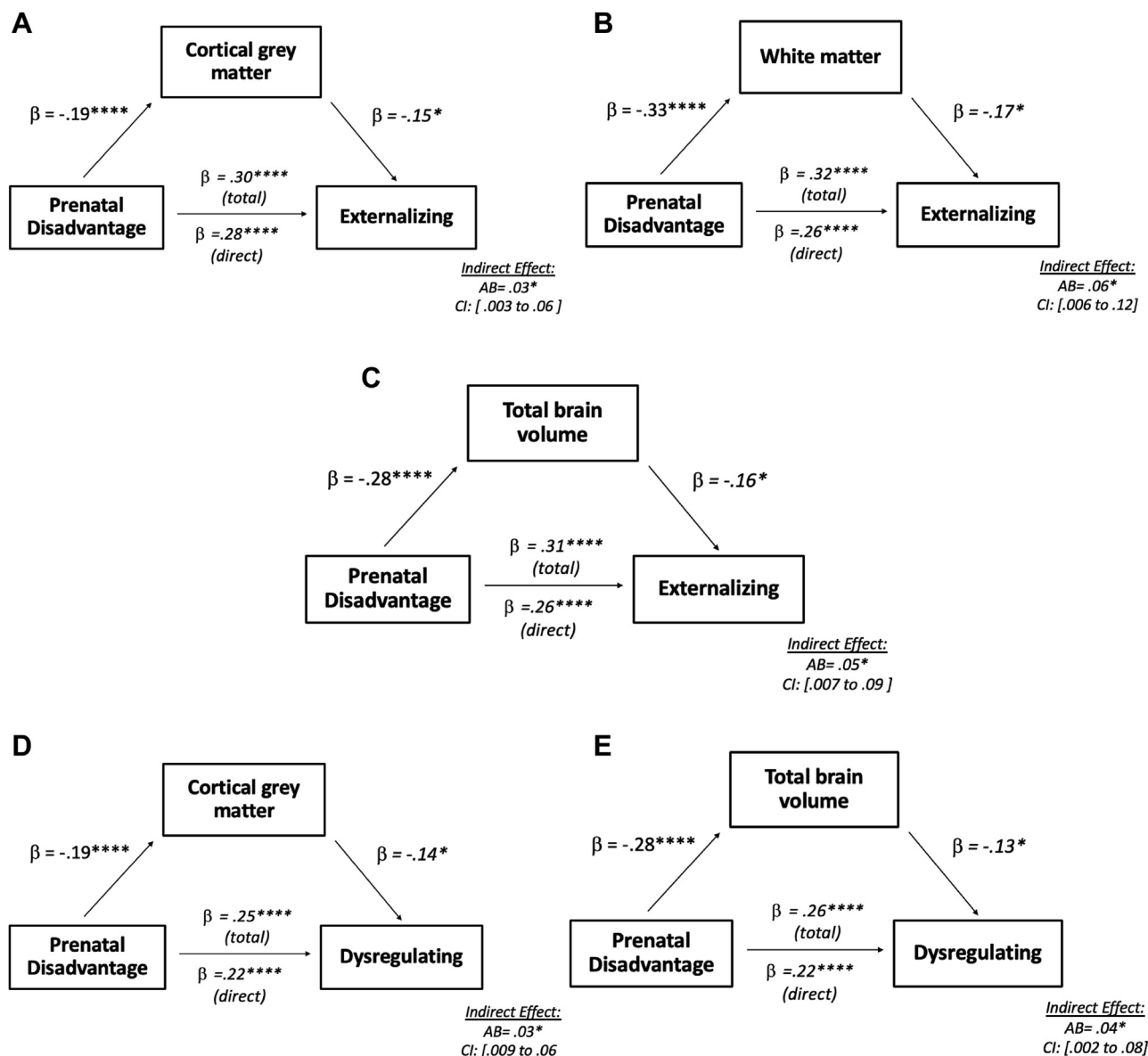


Figure 2. Neonatal brain metrics that partially mediate associations between prenatal disadvantage and socioemotional symptoms at age 2. Smaller cortical gray matter (**A**), white matter (**B**), and total brain (**C**) volumes partially mediate associations between prenatal disadvantage and externalizing symptoms. Smaller cortical gray matter (**D**) and total brain (**E**) volumes partially mediate associations between prenatal disadvantage and dysregulation symptoms. Standardized effect estimates (β) for path A; path B; and total, direct, and indirect effects. Path A describes the effect of the independent variable (prenatal disadvantage) on the mediating variable, controlling for postmenstrual age at scan. Path B describes the unique effect of the mediating variable on the outcome variable, controlling for the independent variable. The direct effect describes the unique effect of the independent variable on the dependent variable after controlling for the mediating variable. The indirect effect is the mediation effect and is displayed with its 95% CI. $^*p < .05$, $^{****}p < .0001$.

supported by multipronged interventions whose targets might shift with time points (e.g., social disadvantage in the prenatal period vs. nonsupportive parenting in early life). These findings are critical given the transdiagnostic value of early socioemotional difficulties in predicting future psychopathology.

PSD-Associated Variability in Socioemotional Abilities Is Evident at Age 2 Years

An existing literature demonstrates social disadvantage-related risk for externalizing and dysregulation symptoms in

school-aged childhood and beyond (10,14,29,46). Our finding that prenatal social disadvantage was associated with higher externalizing and dysregulation symptoms at age 2 extends this literature by demonstrating that these associations are already present in very early life—a developmental period during which these associations have been comparatively underinvestigated. Because early socioemotional difficulties persist throughout childhood and are precursors to future psychopathology (3–7), recognition of the earliest possible risk indicators at the earliest tractable time is critical. As such, this

Table 3. Neonatal Brain Metrics That Do Not Mediate Associations Between Prenatal Disadvantage and Socioemotional Outcomes at Age 2 Years

Outcome Variable	Mediating Variable	β				
		Path A	Path B	Direct Effect	Indirect Effect	Indirect Effect, 95% CI
Externalizing	Subcortical gray matter	−0.25****	−0.09	0.30****	0.02	−0.011 to 0.060
	Cerebellum	−0.18****	−0.07	0.30****	0.01	−0.001 to 0.040
	Mean gyrification index	−0.25****	−0.06	0.21****	0.01	−0.017 to 0.050
Dysregulation	Subcortical gray matter	−0.25****	−0.10	0.23**	−0.03	−0.005 to 0.090
	White matter	−0.33****	−0.11	0.23**	0.04	−0.010 to 0.090
	Cerebellum	−0.25****	−0.09	0.30****	0.02	−0.011 to 0.060

Standardized effect estimates (β s) for path A, path B, direct and indirect effects, and the 95% CIs of the indirect effects. Path A describes the effect of the independent variable (prenatal disadvantage) on the mediating variable, controlling for postmenstrual age at scan. Path B describes the unique effect of the mediating variable on the outcome variable, controlling for the independent variable. The direct effect describes the unique effect of the independent variable on the dependent variable after controlling for the mediating variable. The indirect effect is the mediation effect. No mediations were significant.

** $p < .01$, **** $p < .0001$.

result has several implications for prevention and public policies designed to support optimal socioemotional development. Our findings point to a potential utility in earlier interventions and emphasize the importance of ensuring that pregnant mothers have sufficient levels of resources (57) to support optimal offspring socioemotional development.

Neonatal BVs Mediate Associations Between PSD and Socioemotional Problems Across Domains

Investigations in older samples (21,26) have noted associations between regional and global brain volumetric reductions and various later socioemotional difficulties. Our findings are consistent, but we have demonstrated that these associations

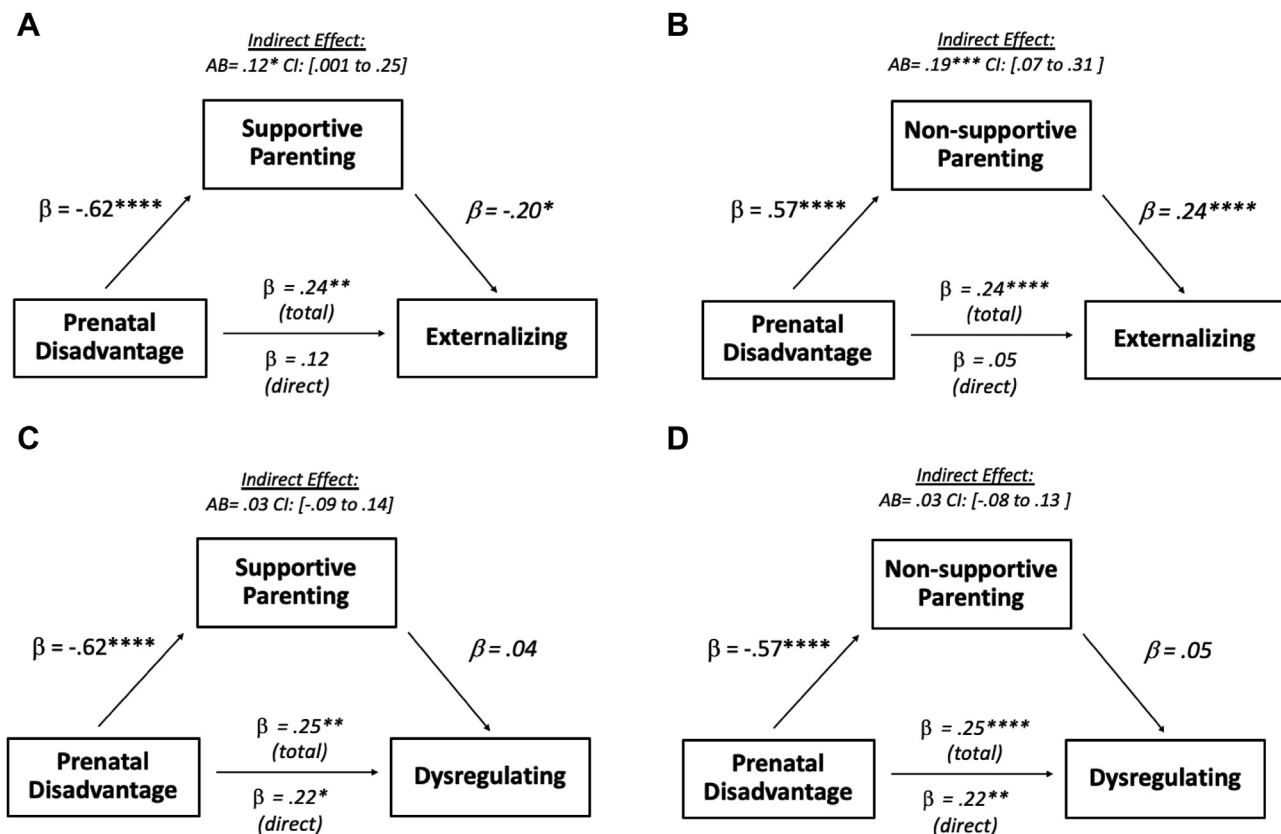


Figure 3. Parenting behaviors at year 1 mediate the relationship of prenatal social disadvantage to externalizing symptoms (A, B) but not to dysregulation symptoms (C, D). Standardized effect estimates (β s) for path A; path B; total, direct, and indirect effects; and 95% CIs of the indirect effects. Path A describes the effect of the independent variable (prenatal disadvantage) on the mediating variable, controlling for postmenstrual age at scan. Path B describes the unique effect of the mediating variable on the outcome variable, controlling for the independent variable. The direct effect describes the unique effect of the independent variable on the dependent variable after controlling for the mediating variable. The indirect effect is the mediation effect and is displayed with its 95% CI. * $p < .05$, ** $p < .01$, *** $p < .001$, **** $p < .0001$.

Table 4. Relative Roles of Parenting and Neonatal Brain Mediators in the Association Between PSD and Externalizing Symptoms

Outcome	Mediating Variables	β		
		Direct Effect of PSD	Indirect Effect	Indirect Effect, 95% CI
Externalizing	Supportive parenting	0.19	0.10	−0.05 to 0.25
	Cortical gray matter volume		0.02	−0.01 to 0.06
	Supportive parenting	0.18	0.09	−0.05 to 0.24
	White matter volume		0.05	−0.004 to 0.10
	Supportive parenting	0.18	0.10	−0.05 to 0.24
	Total brain volume		0.04	−0.007 to 0.09
	Nonsupportive parenting	0.09	0.20 ^a	0.06 to 0.33 ^a
	Cortical gray matter volume		0.02	−0.01 to 0.06
	Nonsupportive parenting	0.07	0.20 ^a	0.06 to 0.33 ^a
	Total white matter volume		0.04	−0.007 to 0.10
	Nonsupportive parenting	0.08	0.20 ^a	0.06 to 0.33 ^a
	Total brain volume		0.04	−0.01 to 0.09

When nonsupportive parenting was entered simultaneously with one of the significant neonatal brain volume mediators in a series of parallel mediation models of the association between prenatal disadvantage and externalizing symptoms, nonsupportive parenting consistently fully mediated the association between PSD and externalizing symptoms while brain metrics did not. Conversely, when nonsupportive parenting was entered simultaneously with one of the significant neonatal brain volume mediators in a series of parallel mediation models of the association between prenatal disadvantage and externalizing symptoms, neither supportive parenting nor neonatal brain volumes mediated the relationship. Postmenstrual age at scan was related to neonatal brain volumes and thus was used as a covariate in path A of the models.

PSD, prenatal social disadvantage.

^aSignificant indirect effect.

are present as early as 2 years of life, which may shift the windows for identification, intervention, and/or prevention even earlier. Our finding that neonatal BVs mediate associations between PSD and both early externalizing and dysregulation behaviors extends current knowledge with a longitudinal demonstration of mediations that had previously only been examined cross-sectionally (28) or later in life (29). This mediation during the neonatal period suggests that alterations in brain structural development in utero are meaningfully related to the development of socioemotional problems. Smaller volumes at birth could indicate delayed trajectories of volumetric growth in utero or predict persistently smaller GM volumes across development (21), which have been associated with increased socioemotional problems (21,58). As such, neonatal structural brain development may 1) serve as a risk factor or biomarker for future risk of early socioemotional problems and associated later psychopathology and 2) as an effective target for prevention (e.g., with programs that support optimal fetal neurodevelopment). This study points to a need to both address social disadvantage during pregnancy as one way to support optimal fetal brain development and to simultaneously identify specific pathways through which PSD is associated with altered neonatal brain development. For example, recent work has suggested that greater interdaily deviations in sleep duration mediated associations between greater PSD and smaller cortical GM volumes in neonates (59) and thus may constitute additional proximal intervention targets.

Parenting Behaviors May Be Particularly Influential in the Development of Externalizing Symptoms

Consistent with an extant literature, we found that when examined separately, both supportive (45,60) and nonsupportive (12,61,62) parenting behaviors mediated associations between disadvantage and externalizing symptoms.

However, when we investigated their simultaneous contributions, only nonsupportive parenting behaviors remained significant mediators. This is consistent with a robust literature (35,43,45) on the equivocal role of supportive behaviors and meta-analytic evidence that nonsupportive behaviors may be more strongly related to risk for externalizing symptoms than low levels of supportive behaviors (35).

Neural and parenting mechanisms are often examined separately; however, investigating these mediators in parallel contributes novel and critical insights about their relative influence on child socioemotional competences. For example, finding that parenting behaviors are particularly powerful mediators may suggest that resources should be devoted to initiatives that support the development of responsive caregiving behaviors, while finding that neonatal brain structure is a salient biomarker may suggest that resources should prioritize supporting healthy pregnancies. When we investigated simultaneous mediations of neonatal BVs and parenting behaviors for PSD-externalizing associations, nonsupportive parenting behaviors remained the sole significant mediator. This suggests that while PSD-associated variability in fetal structural brain development is meaningfully associated with later externalizing symptoms, nonsupportive parenting behaviors may be an even more important mediator. Follow-up analyses with subcomponents of parenting composites demonstrate that detachment was the main driver of these associations (Tables S8–S11). This is consistent with preclinical and human findings that have linked withdrawn parenting behavior to offspring externalizing symptoms and later psychopathology (63,64). This may be because in response to parental disengagement, children increase calling and contact-seeking behavior (63) in bids to garner attention and signal needs, which without contingent parental responses may become increasingly exaggerated and persistent over time, eventually manifesting as externalizing symptoms. Relatedly, detachment may function as an early predictor of

parental monitoring, which also relates to offspring externalizing behaviors. However, relative to other parenting behaviors, detachment has been understudied (64), especially in early life, so future studies should more directly test these hypotheses and pathways, particularly in the context of PSD.

Conversely, the PSD-dysregulation association was mediated by neonatal BVs but not by parenting. Dysregulation, as captured by the ITSEA subscale, may be a more biologically influenced, trait-like socioemotional phenotype. The items in the 2-year ITSEA dysregulation scale assess behaviors ranging from sensory sensitivity, feeding, and sleeping problems to temperament-like items such as negative reactivity and regulation. These items more directly implicate biological systems such as vision, sensation and perception (65), cortisol reactivity (66), and amygdala-cortical connectivity (67). Furthermore, meta-analytic evidence indicates that associations between parenting and dysregulation are very sensitive to the specific type of parenting behavior (e.g., parental regulation) or style (e.g., authoritarian) (68). Thus, one plausible explanation for a lack of mediation by parenting may be that the specific parenting behaviors in our composites may not demonstrate the strongest associations with dysregulation symptoms. In future studies, investigators should consider investigating a broader range of parenting behaviors or including styles.

Limitations and Future Directions

This study has several strengths including a simultaneous examination of neural versus parenting contributions, a longitudinal design, a multidimensional assessment of disadvantage and parenting, and a diverse cohort enriched for exposure to PSD. However, the results should be interpreted in light of several limitations. First, as is common in countries and regions with low economic and social mobility (69), prenatal disadvantage was highly correlated with postnatal disadvantage ($r = 0.92$). Therefore, it was not possible to investigate the degree of specificity in prenatal versus postnatal social disadvantage associations. Future studies with different designs [i.e., interventional, observational in settings with higher social mobility (69)] are needed to disentangle these timing contributions. Similarly, supportive and nonsupportive parenting composites were also highly correlated ($r = -0.76$). Future studies may achieve greater variability in these constructs by using a greater variety of tasks (e.g., natural observations at home, during diaper changes). All components of PSD index socioeconomic deprivation, so future research would benefit from investigating other dimensions of adverse experiences (e.g., threat, unpredictability). Because neonatal BVs and parenting behaviors were assessed at different time points, it is possible that parenting was a stronger mediator because it occurred closer to the outcome time point. Future studies that measure BVs at multiple time points could assess whether time-to-outcome is an important predictor. The magnitude of the bivariate brain-behavior correlations was small, as is typically found. Although significant mediations suggest that neonatal BVs may play an important role in the development of socioemotional symptoms, smaller correlations coupled with partial, rather than full, mediations suggest that additional unmeasured factors may also contribute to behavioral outcomes beyond those captured in this study. Finally, although the ITSEA has been extensively validated in

children 12 to 36 months of age (70–72), it may be difficult for parents to accurately report on children's internalizing symptoms at this age. This may explain the lack of an association between PSD and internalizing symptoms, so future studies should consider the use of observational measures.

Conclusions

Our results demonstrate that PSD is meaningfully related to risk for poorer early socioemotional abilities. Critically, the development of dysregulation and externalizing behaviors are both mediated by PSD-associated neonatal brain structural alterations, suggesting that neonatal brain structure may represent an early risk factor for multidimensional socioemotional difficulties. The development of externalizing abilities was additionally mediated by early parenting behaviors, particularly nonsupportive parenting behaviors, suggesting that such parenting behaviors may serve as a resilience factor in the development of specific socioemotional difficulties (e.g., externalizing problems). As early socioemotional difficulties are strongly associated with later psychopathology, these results are critical in better understanding how we can support optimal trajectories of socioemotional development.

ACKNOWLEDGMENTS AND DISCLOSURES

This work was supported by the National Institutes of Health (NIH) (Grant Nos. R01MH113883 [to JLL, CDS, BW], K01MH122735 [to REL], T32NS121881 [to SL], and T32 MH100019 [to RLT]), the March of Dimes Foundation (Grant No. F30 HD104313 [to RGB]), the Children's Discovery Institute (Grant No. MI-II-2018-725 [to BW]), the Washington University Intellectual and Developmental Disability Research Center (Grant No. P50 HD103525 [to RLT]), NIH Blueprint Diversity Specialized Predoctoral to Postdoctoral Advancement in Neuroscience from the National Institute of Mental Health, (Award No. 5K00MH135485-04 [to BA]), the National Institute of Mental Health, and the American Foundation for Suicide Prevention (Grant Nos. KL2 TR00234 [to RLT] and K23 NS136762 [to RLT]). The funders had no role in the design or conduct of the study; collection, management, analysis, or interpretation of the data; preparation, review, or approval of the article; or the decision to submit the article for publication.

Data are available through the National Data Archive.

The authors report no biomedical financial interests or potential conflicts of interest.

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Received Oct 29, 2024; revised Apr 25, 2025; accepted May 13, 2025.

Supplementary material cited in this article is available online at <https://doi.org/10.1016/j.biopsych.2025.05.015>.

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